

*Answer all questions.*

1. Crude oil can be separated into simpler mixtures, called fractions, which contain hydrocarbon compounds with boiling points in a similar range.

(a) The table lists the properties of some fractions obtained from crude oil.

| Fraction   | Number of carbon atoms in fraction | Boiling point range (°C) | Colour of fraction | Flame when burning | Ease of burning |
|------------|------------------------------------|--------------------------|--------------------|--------------------|-----------------|
| fuel oil   | 1-4                                | –170 to 20               | colourless         | clean              | very easy       |
| petrol     | 5-10                               | 20 to 70                 | pale yellow        | clean              | easy            |
| naphtha    | 8-12                               | 70 to 120                | yellow             | some soot          | quite easy      |
| kerosene   | 10-16                              | 120 to 240               | dark yellow        | smoky              | quite difficult |
| diesel oil | 15-30                              | 240 to 350               | brown              | smoky              | difficult       |

Use the information in the table to describe how the burning of the fractions depends upon the size of the molecules. [2]

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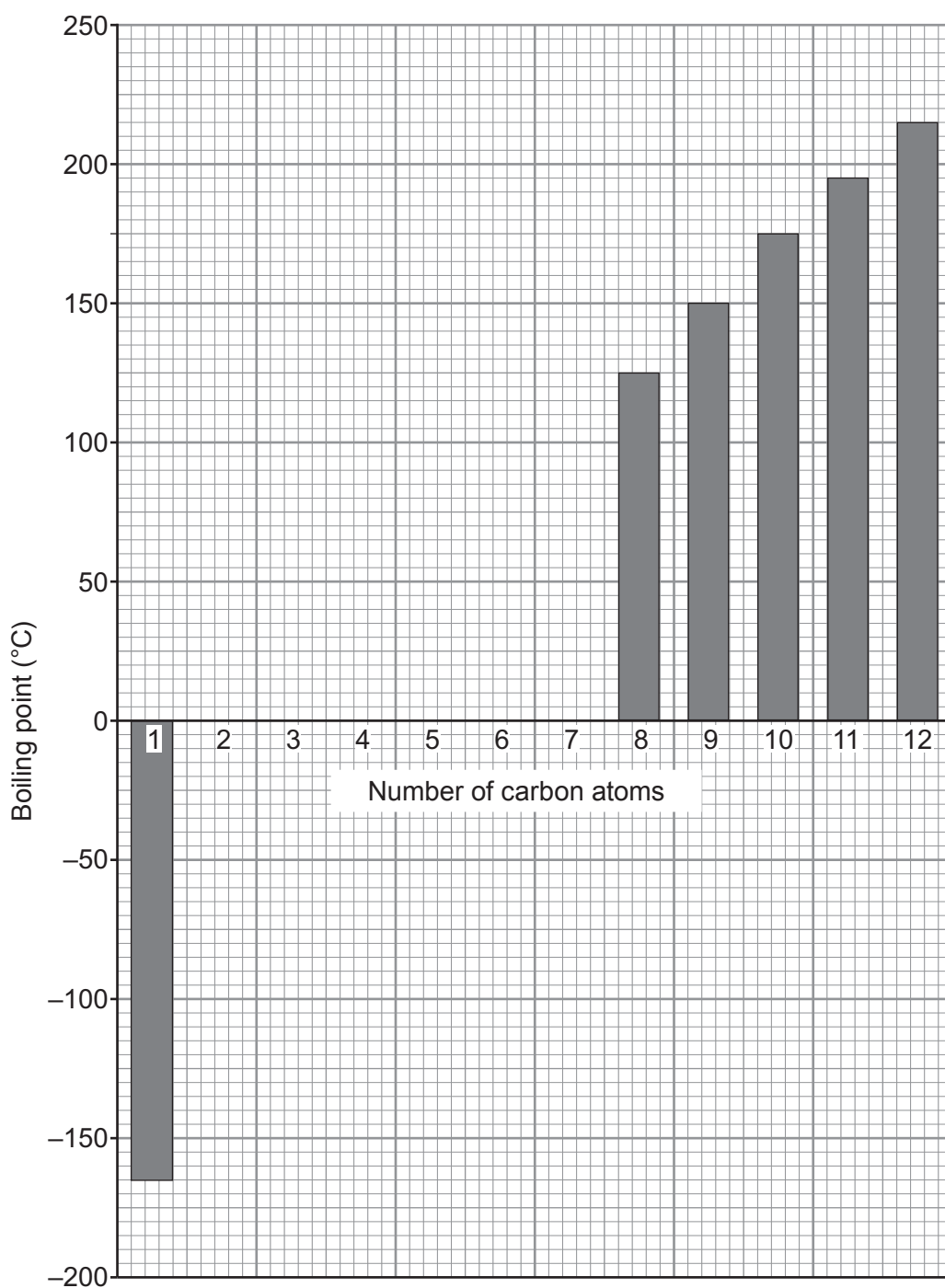
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- (b) The boiling points of hydrocarbons containing 1 to 12 carbon atoms are shown in the table below. The boiling point for the hydrocarbon with 7 carbon atoms is missing.

| Number of carbon atoms | Boiling point (°C) |
|------------------------|--------------------|
| 1                      | –165               |
| 2                      | –90                |
| 3                      | –40                |
| 4                      | 10                 |
| 5                      | 35                 |
| 6                      | 70                 |
| 7                      |                    |
| 8                      | 125                |
| 9                      | 150                |
| 10                     | 175                |
| 11                     | 195                |
| 12                     | 215                |



- (i) Complete the bar chart below. Some of the bars have already been drawn. [2]



- (ii) Use a ruler to draw a trend line **on the chart** and use this to estimate the boiling point of the hydrocarbon with 7 carbon atoms. [2]

Boiling point ..... °C



(c) Many of the fractions obtained from crude oil are used as fuels.

(i) The fire triangle shows the factors necessary to start and maintain a fire.

State **one** method that could be used to safely put out a small amount of spilled petrol burning on the floor. Give the reason why your chosen method would work. [1]

Method .....

Reason .....

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(ii) One of the hydrocarbons in petrol is pentane,  $C_5H_{12}$ .

Complete and balance the symbol equation for the complete combustion of pentane. [2]



(iii) Hydrogen fuel cells are now used in many cars instead of petrol. The overall change inside a hydrogen fuel cell is the same as when hydrogen burns.

Explain why using hydrogen fuel cells in cars is better for the environment than petrol. [2]

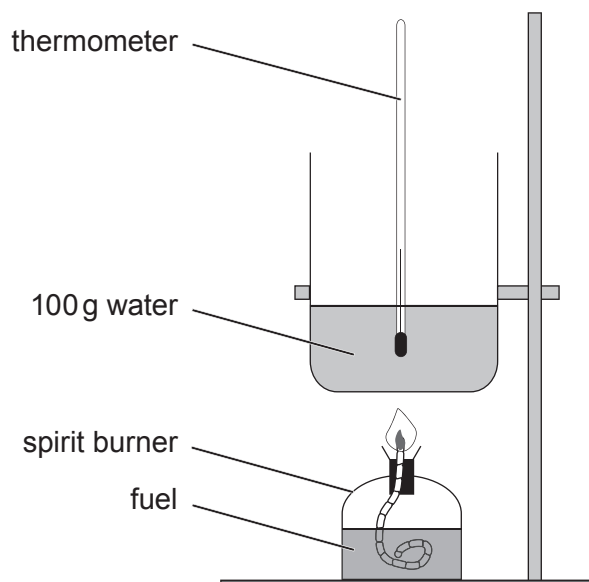
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- (d) It is possible to compare the energy released when different fuels are burned using the following apparatus.



To calculate the energy released per gram of fuel burned, the following equation is used.

$$\text{energy released per gram of fuel (J/g)} = \frac{\text{mass of water} \times 4.2 \times \text{temperature rise (}^{\circ}\text{C)}}{\text{mass of fuel used (g)}}$$

- (i) Apart from measuring the mass of water, describe **all** the measurements that would need to be taken to be able to calculate the energy released per gram of fuel burned. [2]

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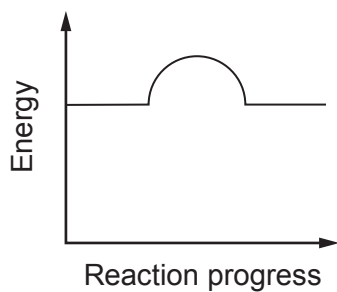
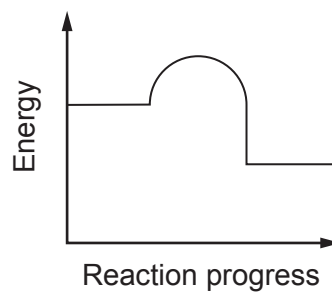
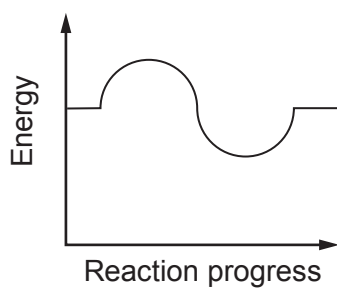
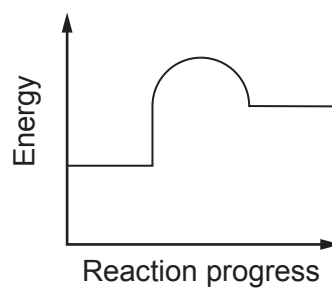
- (ii) When comparing the energy released from different fuels, 100 g of water should be used each time. State **one** other variable that should be controlled. [1]

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(iii) Tick (✓) the energy profile diagram that represents the combustion of fuels. [1]

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